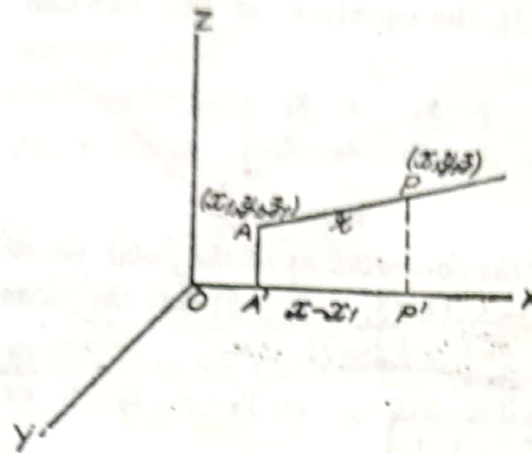


where r is the actual distance of any point on the line from the given point (x_1, y_1, z_1) [l, m, n , being actual d.c.'s].



Thus

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$$

Above are the required equations of a straight line.

Symmetrical form in terms of direction-ratios.

If a, b, c represent the direction-ratios, then the direction cosines are $\frac{a}{\sqrt{(\Sigma a^2)}}, \frac{b}{\sqrt{(\Sigma a^2)}}, \frac{c}{\sqrt{(\Sigma a^2)}}$.

Thus the equations of the line takes the form

$$\frac{x-x_1}{\frac{a}{\sqrt{(\Sigma a^2)}}} = \frac{y-y_1}{\frac{b}{\sqrt{(\Sigma a^2)}}} = \frac{z-z_1}{\frac{c}{\sqrt{(\Sigma a^2)}}}$$

or

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = r \text{ (say).}$$

...(3)

Here r does not represent the distance of any point on the line from (x_1, y_1, z_1) as in the case when we used direction cosines. Hence we observe that the form of equation of the line is the same whether we take direction cosines or direction-ratios.

General co-ordinates of a point on a line whose equation is given in symmetrical form.

From (1) and (2) we find that the co-ordinates of any point on the line are

$$(x_1 + lr, y_1 + mr, z_1 + nr) \text{ or } (x_1 + ar, y_1 + br, z_1 + cr)$$

Cor. To find the equation of a straight line passing through two points.

Let the co-ordinates of the given points be (x_1, y_1, z_1) , and (x_2, y_2, z_2) ; then direction cosines of the line joining these points are proportional to $x_2 - x_1, y_2 - y_1, z_2 - z_1$. [§ 7, P. 19]

Hence from (3), the equation of the line can be put in the form

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

Exercise 8

Ex. 1. Find the co-ordinates of the point where the line joining the points $(2, -3, 1)$, $(3, -4, -5)$ cuts the plane $2x + y + z = 7$.

D.R.'s of the line are $x_2 - x_1, y_2 - y_1, z_2 - z_1$ or $1, -1, 6$. It passes through $(2, -3, 1)$.

Hence the equations are

$$\frac{x-2}{1} = \frac{y+3}{-1} = \frac{z-1}{6} = r \text{ say.}$$

The co-ordinates of any point on this line are $(r+2, -r-3, 6r+1)$.

If the point lies on the plane $2x + y + z = 7$, we have $2r+4-r-3-6r+1=7$; $\therefore r=-1$.

Putting the value of r we get the co-ordinates of the required point as $(1, -2, 7)$.

(b) Prove that the line $\frac{x-1}{2} = \frac{2-y}{3} = \frac{z+3}{4}$ meets the plane $2x+4y-z-1=0$ in the point $(3, -1, 1)$. [Vikram 78; Meerut 75]

Ex. 2. (a) Find the equations of the straight line through (a, b, c) , which are (i) parallel to z -axis (i.e.) perpendicular to $(xy$ -plane), (ii) perpendicular to z -axis (i.e.) parallel to xy -plane.

[Berhampur 81; Indore 82]

Any line through (a, b, c) is $\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{n}$.

If it is parallel to z -axis or (perp. to xy -plane), whose d.c.'s are $(0, 0, 1)$, then l, m, n are proportional to these numbers and hence the equations are $\frac{x-a}{0} = \frac{y-b}{0} = \frac{z-c}{1}$.

If it is perpendicular to z -axis or (parallel to xy -plane), then $l \cdot 0 + m \cdot 0 + n \cdot 1 = 0$.

$\therefore n=0$ and the equations are $\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{0}$.

(b) Find the equations to the line joining $(1, 2, 3)$ and $(-3, 4, 3)$; what conclusion do you draw as regards its characteristics?

Ans. $(x-1)/4 = (y-2)/-2 = (z-3)/0$; perpendicular to z -axis, i.e. parallel to xy -plane by part (a).

Ex. 3 (a) Prove that the distance of the point of intersection of the line

$$\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$$

and the plane $x-y+z=5$ from the point $(-1, -5, -10)$ is 13.

[Avadh 81; Agra 80; Calicut 82; Vikram 78; Madras 75]

Any point on the line is $3r+2, 4r-1, 12r+2$.

If it lies on the plane $x-y+z=5$, then

$$(3r+2) - (4r-1) + (12r+2) = 5. \quad \therefore r=0$$

Hence the point of intersection is $(2, -1, 2)$. Its distance from $(-1, -5, -10)$ is

$$\sqrt{[(2+1)^2 + (-1+5)^2 + (2+10)^2]} = \sqrt{(9+16+144)} = 13.$$

(b) Prove that the distance of the point of intersection of the line $\frac{x-3}{1} = \frac{y-4}{2} = \frac{z-5}{2}$ and the plane $x+y+z=2$ from the point $(3, 4, 5)$ is 6. [Jiwaji 63]

Point of intersection is $(1, 0, 1), r=-2$.

Ex. 4. Find the points in which the line

$$\frac{x+1}{-1} = \frac{y-12}{5} = \frac{z-7}{2}$$

cuts the surface $11x^2 - 5y^2 + z^2 = 0$.

Any point on the line is $-r-1, 5r+12, 2r+7$.

If it lies on the given surface then

$$11(-r-1)^2 - 5(5r+12)^2 + (2r+7)^2 = 0$$

$$\text{or } 110r^2 + 550r + 660 = 0 \text{ or } r^2 + 5r + 6 = 0; \quad \therefore r = -3, -2.$$

Putting for r the points of intersection are

$$(2, -3, 1) \text{ and } (1, 2, 3).$$

Ex. 5. Find the distance of point $(1, -2, 3)$ from the plane $x-y+z=5$ measured parallel to the line whose direction cosines are proportional to $2, 3, -6$.

[Agra 81, L.N.M. 82, Rewa 71; Bhopal 71]

Here we are not required to find the perpendicular distance of the point $(1, -2, 3)$ from the plane, but the distance measured

same D.R.'s are proportional to a, b, c is perpendicular to the given plane.

$$\therefore (a_1 + 2a_2 + 3a_3 = 0 \text{ and } (b_1 + 2b_2 + 3b_3 = 0$$

$$\therefore \frac{a_1}{a} = \frac{a_2}{a} = \frac{a_3}{a} = \frac{b_1}{b} = \frac{b_2}{b} = \frac{b_3}{b} = \frac{c_1}{c} = \frac{c_2}{c} = \frac{c_3}{c}$$

Putting for a, b, c in (1), we get the plane as

$$x(x-x_1) + m(y-y_1) + n(z-z_1) = 0$$

(ii) Since the plane is perpendicular to the line therefore normal is parallel to the line and hence d.c.'s are same i.e. l, m, n .

$$\therefore \text{Plane is } l(x-x_1) + m(y-y_1) + n(z-z_1) = 0.$$

Foot of perpendicular from a point to a plane. [Image]

Ex. 8. Show that if the axes are rectangular, the equation to the perpendicular from the point (α, β, γ) to the plane $ax + by + cz + d = 0$

are $\frac{x-\alpha}{a} = \frac{y-\beta}{b} = \frac{z-\gamma}{c}$ and deduce the perpendicular distance of the point (α, β, γ) from the plane. Find also the co-ordinates of the foot of the perpendicular.

The line being perpendicular to plane, i.e. parallel to normal so that its d.c.'s are proportional to a, b, c the D.R.'s of the normal to the plane. Also it passes through (α, β, γ) hence its equations are

$$\frac{x-\alpha}{a} = \frac{y-\beta}{b} = \frac{z-\gamma}{c} = r \text{ (say).}$$

Any point on this line is

$$(ar + \alpha, br + \beta, cr + \gamma)$$

...(1)

If it lies on the plane $ax + by + cz + d = 0$, then

$$a(ar + \alpha) + b(br + \beta) + c(cr + \gamma) + d = 0$$

$$\text{or } r = -\frac{a\alpha + b\beta + c\gamma + d}{(a^2 + b^2 + c^2)} \quad \dots(2)$$

Putting the value of r in (1), we shall find the co-ordinates of the foot of the perpendicular. This is called the image of the given point in the given plane.

Now perp. distance of plane from (α, β, γ) is the distance between points (α, β, γ) and $(ar + \alpha, br + \beta, cr + \gamma)$ for value of r given by (2) and is

$$\begin{aligned} \sqrt{(a^2r^2 + b^2r^2 + c^2r^2)} &= r \cdot \sqrt{(a^2 + b^2 + c^2)} \\ &= -\frac{a\alpha + b\beta + c\gamma + d}{\sqrt{(a^2 + b^2 + c^2)}} \text{ by (2)} = \frac{a\alpha + b\beta + c\gamma + d}{\sqrt{(a^2 + b^2 + c^2)}} \text{ in magnitude} \end{aligned}$$

(b) Find the image of the point $(2, -1, 3)$ in the plane

$$3x - 2y + z = 9.$$

[Agra 80]

parallel to line whose D.R.'s are 2, 3, -6. Any line through (1, -2, 3) parallel to the line whose D.R.'s are 2, 3, -6 is

$$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-6} = r \text{ say}$$

Any point on it is $(2r+1, 3r-2, -6r+3)$

If it lies on the plane $x-y+z=5$, then

$$(2r+1) - (3r-2) + (-6r+3) = 5. \quad \therefore r = \frac{1}{7}$$

$\therefore$ point is $\left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$ and its distance from (1, -2, 3) is

$$\frac{1}{7} \sqrt{(4+9+36)} = \frac{7}{7} = 1.$$

Ex. 6. (a) Prove that the equations of the line through (x_1, y_1, z_1) perpendicular to the lines

$$\frac{x}{l_1} = \frac{y}{m_1} = \frac{z}{n_1} \text{ and } \frac{x}{l_2} = \frac{y}{m_2} = \frac{z}{n_2}$$

and
$$\frac{x-x_1}{m_1n_2-m_2n_1} = \frac{y-y_1}{n_1l_2-n_2l_1} = \frac{z-z_1}{l_1m_2-l_2m_1}$$

[Luck. 89, Jodhpur 78, Pb. 60]

Let the line through (x_1, y_1, z_1) be

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}.$$

Since it is perpendicular to the given lines,

$$\therefore ll_1 + mm_1 + nn_1 = 0, ll_2 + mm_2 + nn_2 = 0.$$

$$\frac{l}{m_1n_2-m_2n_1} = \frac{m}{n_1l_2-n_2l_1} = \frac{n}{l_1m_2-l_2m_1}$$

Putting the proportionate values of l, m, n in (1), we get the required equation as given. ✓

Ex. 7. Find the equations of the plane through the point (x, y, z) and (i) parallel to the lines

$$\frac{x}{l_1} = \frac{y}{m_1} = \frac{z}{n_1} \text{ and } \frac{x}{l_2} = \frac{y}{m_2} = \frac{z}{n_2};$$

(ii) perpendicular to the straight line

$$\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$$

[Osmania 83]

(i) Any plane through (x_1, y_1, z_1) is

$$A(x-x_1) + B(y-y_1) + C(z-z_1) = 0$$

Since the plane is parallel to the lines, hence the normal

*

whose D.C.'s are proportional to A, B, C is perpendicular to the given lines

$$\therefore Al_1 + Bm_1 + Cn_1 = 0 \text{ and } Al_2 + Bm_2 + Cn_2 = 0$$

$$\therefore \frac{A}{m_1n_2 - m_2n_1} = \frac{B}{n_1l_2 - n_2l_1} = \frac{C}{l_1m_2 - l_2m_1}$$

Putting for A, B, C in (1), we get the plane as

$$\Sigma (x - x_1) (m_1n_2 - m_2n_1) = 0$$

(ii) Since the plane is perpendicular to the line therefore normal is parallel to the line and hence d.c.'s are same i.e. l, m, n .

$$\therefore \text{Plane is } l(x - x_1) + m(y - y_1) + n(z - z_1) = 0.$$

Foot of perpendicular from a point to a plane. [Image]

Ex. 8. Show that if the axes are rectangular, the equation to the perpendicular from the point (α, β, γ) to the plane $ax + by + cz + d = 0$

are $\frac{x - \alpha}{a} = \frac{y - \beta}{b} = \frac{z - \gamma}{c}$ and deduce the perpendicular distance of the point (α, β, γ) from the plane. Find also the co-ordinates of the foot of the perpendicular.

The line being perpendicular to plane, i.e. parallel to normal so that its d.c.'s are proportional to a, b, c the D.R.'s of the normal to the plane. Also it passes through (α, β, γ) hence its equations are

$$\frac{x - \alpha}{a} = \frac{y - \beta}{b} = \frac{z - \gamma}{c} = r \text{ (say).}$$

Any point on this line is

$$(ar + \alpha, br + \beta, cr + \gamma) \quad \dots(1)$$

If it lies on the plane $ax + by + cz + d = 0$, then

$$a(ar + \alpha) + b(br + \beta) + c(cr + \gamma) + d = 0$$

$$\text{or } r = -\frac{a\alpha + b\beta + c\gamma + d}{(a^2 + b^2 + c^2)} \quad \dots(2)$$

Putting the value of r in (1), we shall find the co-ordinates of the foot of the perpendicular. This is called the image of the given point in the given plane.

Now perp. distance of plane from (α, β, γ) is the distance between points (α, β, γ) and $(ar + \alpha, br + \beta, cr + \gamma)$ for value of r given by (2) and is

$$\begin{aligned} & \sqrt{(a^2r^2 + b^2r^2 + c^2r^2)} = r \cdot \sqrt{(a^2 + b^2 + c^2)} \\ & = -\frac{a\alpha + b\beta + c\gamma + d}{\sqrt{(a^2 + b^2 + c^2)}} \text{ by (2)} = \frac{a\alpha + b\beta + c\gamma + d}{\sqrt{(a^2 + b^2 + c^2)}} \text{ in magnitude} \end{aligned}$$

(b) Find the image of the point $(2, -1, 3)$ in the plane

$$3x - 2y + z = 9.$$

[Agra 80]

As in part (a) the line through $(2, -1, 3)$ perpendicular to given plane is

$$\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z-3}{1} = r \text{ say}$$

Its intersection with the plane is found as usual to be $(\frac{22}{7}, -\frac{8}{7}, \frac{20}{7})$ which is the image of given point.

(c) Prove that the image of the point $(1, 3, 4)$ in the plane

$$2x - y + z + 3 = 0 \text{ is } (-1, 4, 3).$$

[Madras 76]

Ex. 9. Find the co-ordinates of the foot of the perpendicular drawn from the origin to the plane $2x + 3y - 4z + 1 = 0$. Find also the co-ordinates of the point on the line which is at the same distance from the foot of the perpendicular as the origin is

[Punjab 75]

Line through origin perp. to plane $2x + 3y - 4z + 1 = 0$, i.e. parallel to its normal whose D.R.'s are 2, 3, -4 is $\frac{x}{2} = \frac{y}{3} = \frac{z}{-4} = r$. Any point on it is $2r, 3r, -4r$. It will lie on the plane if $4r + 9r + 16r + 1 = 0$; $\therefore r = -\frac{1}{29}$.

The co-ordinates of the foot, of the perpendicular P are $(-\frac{2}{29}, -\frac{3}{29}, \frac{4}{29})$. If O' be any other point on the line such that $OP = O'P$, then P is the mid-point of OO' . O is $(0, 0, 0)$ and suppose O' is (x, y, z) ; then

$$\frac{x+0}{2} = -\frac{2}{29}, \frac{y+0}{2} = -\frac{3}{29}, \frac{z+0}{2} = \frac{4}{29}$$

$$\therefore O' \text{ is } \left(-\frac{4}{29}, -\frac{6}{29}, \frac{8}{29}\right)$$

Ex. 10. A variable plane makes intercepts on the co-ordinate axes the sum of whose squares is constant and equal to k^2 . Show that the locus of the foot of the perpendicular from origin to the plane is

$$(x^2 + y^2 + z^2)(x^2 + y^2 + z^2)^2 = k^2 \quad [\text{Burdwan Hons. 75}]$$

$$\text{Let the plane be } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

...(1)

where by given condition $a^2 + b^2 + c^2 = k^2$.

...(2)

Any line through origin and perpendicular to plane i.e. parallel to normal to the plane whose D.R.'s are

$$\frac{1}{a}, \frac{1}{b}, \frac{1}{c} \text{ is } \frac{x-0}{1/a} = \frac{y-0}{1/b} = \frac{z-0}{1/c} = r.$$

...(3)

$$\text{Any point on it is } \left(\frac{r}{a}, \frac{r}{b}, \frac{r}{c}\right).$$

...(4)

It will lie on the plane (1) if

$$r \left(\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) = 1; \therefore r = \frac{1}{a^{-2} + b^{-2} + c^{-2}}$$

Putting the values of r in (4), the co-ordinates of foot of perpendicular are

$$x = \frac{a^{-1}}{a^{-2} + b^{-2} + c^{-2}}, y = \frac{b^{-1}}{\Sigma a^{-2}}, z = \frac{c^{-1}}{\Sigma a^{-2}}$$

$$\therefore x^2 + y^2 + z^2 = \frac{a^{-2} + b^{-2} + c^{-2}}{(a^{-2} + b^{-2} + c^{-2})^2} = \frac{1}{(a^{-2} + b^{-2} + c^{-2})} \quad \dots(5)$$

$$\text{Also } x^{-2} + y^{-2} + z^{-2} = (a^{-2} + b^{-2} + c^{-2})^2 (a^2 + b^2 + c^2)$$

$$= \frac{1}{(x^2 + y^2 + z^2)^2} \cdot k^2 \text{ by (2) and (5)}$$

$$\therefore \text{Locus is } (x^{-2} + y^{-2} + z^{-2}) (x^2 + y^2 + z^2)^2 = k^2.$$

Ex. 11. Find the equation of the line through the points (a, b, c) and (a', b', c') and prove that it passes through the origin if $aa' + bb' + cc' = \rho\rho'$ where ρ and ρ' are the distances of the points from the origin.

[Allahabad 77; Indore 76; Kanpur 76]

$$\frac{x-a}{a'-a} = \frac{y-b}{b'-b} = \frac{z-c}{c'-c}. \text{ It passes through origin}$$

$$\therefore \frac{0-a}{a'-a} = \frac{0-b}{b'-b} = \frac{0-c}{c'-c} \text{ or } \frac{a'-a}{-a} = \frac{b'-b}{-b} = \frac{c'-c}{-c}$$

$$\text{or } \frac{-a'}{a} + 1 = \frac{-b'}{b} + 1 = \frac{-c'}{c} + 1$$

$$\text{or } \frac{a'}{a} = \frac{b'}{b} = \frac{c'}{c}$$

$$\therefore ab' - a'b = 0, bc' - b'c = 0, ca' - c'a = 0. \quad \dots(1)$$

Now by Lagrange's identity, we know that

$$(a^2 + b^2 + c^2)(a'^2 + b'^2 + c'^2) - (aa' + bb' + cc')^2 = (ab' - a'b)^2 + (bc' - b'c)^2 + (ca' - c'a)^2 = 0 \text{ from (1).} \quad \dots(2)$$

Now ρ and ρ' are the distances of the points (a, b, c) and (a', b', c') respectively from origin.

$$\therefore \rho^2 \rho'^2 = (a^2 + b^2 + c^2)(a'^2 + b'^2 + c'^2) = (aa' + bb' + cc')^2 \text{ by (2)}$$

or $aa' + bb' + cc' = \rho\rho'.$

Ex. 12. If the plane $lx + my + nz = p$ where $l^2 + m^2 + n^2 = 1$ meets the co-ordinate axes in X, Y, Z and G is the centroid of the triangle XYZ and if perpendicular to the plane at G , meets the co-ordinate planes in A, B, C then prove that

$$\frac{1}{GA} + \frac{1}{GB} + \frac{1}{GC} = \frac{3}{p}$$

[Karnatak 69]

The points X, Y, Z are $\left(\frac{p}{l}, 0, 0\right), \left(0, \frac{p}{m}, 0\right), \left(0, 0, \frac{p}{n}\right)$.

$\therefore G$ the centroid of triangle XYZ is

$$\left(\frac{p}{3l}, \frac{p}{3m}, \frac{p}{3n}\right).$$

Any line through G and perpendicular to the plane $lx+my+nz=p$ or parallel to its normal whose d.c.'s are l, m, n is

$$\frac{x-\frac{p}{3l}}{l} = \frac{y-\frac{p}{3m}}{m} = \frac{z-\frac{p}{3n}}{n} = r$$

where r is the actual distance of G from any point of the line.

This line meets the plane $x=0$ in A ;

$\therefore$ putting $x=0$ and $r=GA$ in the above, we get

$$\frac{0-\frac{p}{3l}}{l} = GA; \quad \therefore \frac{-p}{3l^2} = GA$$

or

$$\frac{1}{GA} = \frac{-3}{p} l^2.$$

Similarly $\frac{1}{GB} = \frac{-3}{p} m^2, \frac{1}{GC} = \frac{-3}{p} n^2.$

$$\therefore \frac{1}{GA} + \frac{1}{GB} + \frac{1}{GC} = \frac{-3}{p} \Sigma l^2 = \frac{-3}{p}.$$

Ex. 13. From a point P , whose co-ordinates are (x, y, z) the perpendicular PM is drawn to the straight line through the origin whose direction-cosines are l, m, n and is produced to P' where $PM=MP'$. If the co-ordinates of P' are (x', y', z') show that

$$\frac{x+x'}{l} = \frac{y+y'}{m} = \frac{z+z'}{n} = (lx+my+nz). \quad [\text{Raj. 74; Punjab 72}]$$

M the mid point of P and P' is

$$\frac{x+x'}{2}, \frac{y+y'}{2}, \frac{z+z'}{2} \text{ and it lies on } \frac{x}{l} = \frac{y}{m} = \frac{z}{n}$$

$$\therefore \frac{x+x'}{l} = \frac{y+y'}{m} = \frac{z+z'}{n} = K \text{ say} \quad \dots(1)$$

or

$$\begin{aligned} \frac{l(x+x')}{l^2} &= \frac{m(y+y')}{m^2} = \frac{n(z+z')}{n^2} \\ &= \frac{\Sigma l(x+x')}{\Sigma l^2} = \frac{\Sigma l(x+x')}{1} = K \quad \dots(2) \end{aligned}$$

Now PP' is perpendicular to the line with d.c.'s l, m, n

$$\therefore l(x-x') + m(y-y') + n(z-z') = 0.$$

$$\therefore lx + my + nz = lx' + my' + nz' \text{ or } \Sigma lx = \Sigma lx' \quad \dots(3)$$

$\therefore K = \Sigma l(x + x') = 2\Sigma lx$ from (2) and (3). Hence from (1), we get the required result.

§ 3. To transform the general form of the equation of a straight line to symmetrical form.

Let the general form of the equation of the straight line be given by

$$\left. \begin{aligned} a_1x + b_1y + c_1z + d_1 &= 0, \\ a_2x + b_2y + c_2z + d_2 &= 0 \end{aligned} \right\} \quad \dots(1)$$

In order to write down the corresponding symmetric form we must know the direction cosines or direction-ratios of the line and at the same time co-ordinates of a point on it.

Let l, m, n be the direction-cosines or direction-ratios of the line. Since the line is common to both the planes, their normals are perpendicular to it. The direction-ratios of the normals are the coefficients of x, y and z in the equations of the planes.

$$\therefore a_1l + b_1m + c_1n = 0 \text{ and } a_2l + b_2m + c_2n = 0.$$

$$\therefore \frac{l}{b_1c_2 - b_2c_1} = \frac{m}{c_1a_2 - c_2a_1} = \frac{n}{a_1b_2 - a_2b_1}$$

Hence the direction-cosines of the line are proportional to

$$b_1c_2 - b_2c_1, c_1a_2 - c_2a_1, a_1b_2 - a_2b_1. \quad \dots(2)$$

In order to find the co-ordinates of any point on the line we can choose it in a number of ways. For the sake of convenience let us choose the point where this line cuts the xy plane i.e. where $z=0$. Putting $z=0$ in (1), we get

$$a_1x + b_1y + d_1 = 0, \quad a_2x + b_2y + d_2 = 0.$$

$$\therefore \frac{x}{b_1d_2 - b_2d_1} = \frac{y}{d_1a_2 - d_2a_1} = \frac{1}{a_1b_2 - a_2b_1}$$

Hence the co-ordinates of required point are

$$x = \frac{b_1d_2 - b_2d_1}{a_1b_2 - a_2b_1}, \quad y = \frac{d_1a_2 - d_2a_1}{a_1b_2 - a_2b_1} \text{ and } z=0. \quad \dots(3)$$

In case $a_1b_2 - a_2b_1 = 0$ then instead of putting $z=0$, we put $x=0$ or $y=0$.

Having found the direction-ratios and the co-ordinates of a point on the line, its equation can be put in the symmetrical form as

$$\frac{x - \frac{b_1d_2 - b_2d_1}{a_1b_2 - a_2b_1}}{b_1c_2 - b_2c_1} = \frac{y - \frac{d_1a_2 - d_2a_1}{a_1b_2 - a_2b_1}}{c_1a_2 - c_2a_1} = \frac{z - 0}{a_1b_2 - a_2b_1}$$

Ex. 1. Find in symmetrical form the equation of a line $x+y+z+1=0$, $4x+y-2z+2=0$ and find its direction-cosines.

[Punjab 73]

If l, m, n be the direction-cosines, then

$$l+m+n=0, 4l+m-2n=0.$$

$$\therefore \frac{l}{-3} = \frac{m}{6} = \frac{n}{-3}$$

or $\frac{l}{-1} = \frac{m}{2} = \frac{n}{-1} = \frac{\sqrt{(l^2+m^2+n^2)}}{\sqrt{(1+4+1)}} = \frac{1}{\sqrt{6}}$

and direction-ratios are $-1, 2, -1$.

Putting $z=0$ we have $x+y+1=0$, $4x+y+2=0$.

$$\therefore \frac{x}{1} = \frac{y}{2} = \frac{1}{-3}; \therefore x = -\frac{1}{3}, y = -\frac{2}{3}$$

$$\therefore \frac{x+\frac{1}{3}}{-1} = \frac{y+\frac{2}{3}}{2} = \frac{z}{-1} \text{ are the required equations.}$$

(b) Prove that the equations $4x+4y-5z=12$ and $8x+12y-13z=32$ represent the line whose equations are

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z}{4}.$$

§ 4. To find the co-ordinates of the point of intersection of a given line and a given plane and deduce the condition that

(a) the line may be parallel to the plane.

[Bhopal 71 ; Utkal 69]

(b) the line may be perpendicular to the plane.

(c) the line may be lying in the plane.

[Garhwal 75; Sambalpur 73; Madras 74, 73; Mysore 73; 70;

Ravishanker 71; Madurai 70; Vikram 73, 71; Jiwaji 66;

Agra 59; Punjab 69; Delhi Hons. 51; Kerala 66, 64;

Sagar B. Sc. 66]

Let the equations of the line in symmetrical form be

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$$

and the equation of the plane be $ax+by+cz+d=0$ (2)

Any point on the line is (x_1+lr, y_1+mr, z_1+nr) , ... (3)

If the point lies on the plane, then

$$a(x_1+lr)+b(y_1+mr)+c(z_1+nr)+d=0$$

The above article (4) is very important and we shall very frequently use it. Students should clearly understand it.

or
$$r = -\frac{ax_1 + by_1 + cz_1 + d}{al + bm + cn}$$

(a) Condition of parallelism.

$$al + bm + cn = 0 \text{ and } ax_1 + by_1 + cz_1 + d \neq 0.$$

If the line is parallel to the plane, its normal is perpendicular to the line and hence we have $al + bm + cn = 0$. The second condition simply emphasizes that the point (x_1, y_1, z_1) does not lie on the plane for otherwise the line will lie in the plane.

Otherwise also we know that if the line be parallel to the plane it will meet it at infinity and hence the value of r obtained from (4) should be infinite the condition for which is that $al + bm + cn = 0$ and at the same time $ax_1 + by_1 + cz_1 + d$ is not equal to zero.

(b) Condition of Perpendicularity

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$$

If the line is perpendicular to the plane, its normal is parallel to it and hence we have $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$.

(c) Condition that the line may lie in the plane

$$al + bm + cn = 0, ax_1 + by_1 + cz_1 + d = 0.$$

[Avadh '80; Calicut '84; Magadh '76]

If the line lies in the plane, then for all values of r , the point $(x_1 + lr, y_1 + mr, z_1 + nr)$ should lie on the plane

$$ax + by + cz + d = 0$$

or
$$r(al + bm + cn) + ax_1 + by_1 + cz_1 + d = 0$$

for all values of r , the conditions for which are

$$al + bm + cn = 0 \text{ and } ax_1 + by_1 + cz_1 + d = 0.$$

Geometrically.

Otherwise also, if the line lies in the plane, then the normal being perpendicular to it we have $al + bm + cn = 0$ and the point (x_1, y_1, z_1) through which the line passes also lies in the plane and hence $ax_1 + by_1 + cz_1 + d = 0$.

Exercise 9

Ex. 1. Find the equation of the line through (x_1, y_1, z_1) parallel to the line

$$l_1x + m_1y + n_1z + p_1 = 0, l_2x + m_2y + n_2z + p_2 = 0. \quad [\text{Ranchi '75}]$$

If l, m, n be the d.c.'s of the line, then it is perpendicular to normal of the planes.

$$\therefore \frac{l_1 + mm_1 + nn_1}{1} = 0, \quad \frac{l_2 + mm_2 + nn_2}{1} = 0;$$

$$\therefore \frac{l_1}{m_1 n_2 - m_2 n_1} = \frac{m}{n_1 l_2 - n_2 l_1} = \frac{n}{l_1 m_2 - l_2 m_1}$$

Also it passes through (x_1, y_1, z_1) ; $\therefore$ its equations are

$$\frac{x-x_1}{m_1 n_2 - m_2 n_1} = \frac{y-y_1}{n_1 l_2 - n_2 l_1} = \frac{z-z_1}{l_1 m_2 - l_2 m_1}$$

Ex. 2. Find the equation of the line through the point $(1, 2, 3)$ parallel to the line $x - y + 2z = 5, 3x + y + z = 6$. [Magadh 76, Ranchi 79]

Proceed as above.

$$\text{Ans. } \frac{x-1}{-3} = \frac{y-2}{5} = \frac{z-3}{4}$$

Ex. 3. Find the direction-cosines of the line whose equations are $x + y = 3$ and $x + y + z = 0$ and show that it makes an angle of 30° with the plane $y - z + 2 = 0$.

Let l, m, n be the d.c.'s of the line.

$$\therefore l + m + 0n = 0.$$

$$\text{and } l + m + n = 0$$

$$\therefore \frac{l}{1} = \frac{m}{-1} = \frac{n}{0}; \therefore \text{D.R.'s are } 1, -1, 0.$$

$$\therefore \text{D.C.'s are } \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0 \right)$$

Again D.R.'s of normal to the plane, $y - z + 2 = 0$ are $0, 1, -1$ or d.c.'s are $0, \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$.

If θ be the angle between the line and plane, then $90 - \theta$ will be the angle between line and normal to the plane.

$$\therefore \cos(90 - \theta) = \frac{1}{\sqrt{2}} \cdot 0 - \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} + 0 = -\frac{1}{2}$$

$$\text{or } \sin \theta = -\frac{1}{2} \text{ or } \theta = 210^\circ = 180^\circ + 30^\circ.$$

or we can also say that $\theta = 30^\circ$.

Ex. 4. Prove that two lines in which the planes $3x - 7y - 5z = 1$ and $5x - 13y + 3z + 2 = 0$ cut the plane $8x - 11y + 2z = 0$ include a right angle.

If l, m, n be the d.c.'s of the line given by

$$3x - 7y - 5z = 1 \text{ and } 8x - 11y + 2z = 0$$

then it is perpendicular to normals of both the planes

$$\therefore 3l - 7m - 5n = 0$$

$$8l - 11m + 2n = 0.$$

*

$$\therefore \frac{l}{-69} = \frac{m}{-46} = \frac{n}{23}$$

$\therefore$ D.R.'s are 3, 2, -1.

The D.R.'s of the line given by

$5x-13y+3z+2=0$ and $8x-11y+2z=0$ are given by

$$5l-13m+3n=0$$

$$8l-11m+2n=0$$

$$\therefore \frac{l}{7} = \frac{m}{14} = \frac{n}{49}$$

$\therefore$ D.R.'s are 1, 2, 7.

Again $a_1a_2+b_1b_2+c_1c_2=3 \cdot 1+2 \cdot 2+(-1) \cdot 7=0$.

Hence the lines are perpendicular.

Ex. 5. Find the equation of the plane through the line $3x-4y+5z=10$, $2x+2y-3z=4$ and parallel to $x=2y=3z$.

Any plane through given line is $P+\lambda Q=0$

$$\text{or } (3x-4y+5z-10)+\lambda(2x+2y-3z-4)=0$$

$$\text{or } (3+2\lambda)x+(-4+2\lambda)y+(5-3\lambda)z-(10+4\lambda)=0 \quad \dots(1)$$

It is parallel to $x=2y=3z$ or $\frac{x}{1}=\frac{y}{\frac{1}{2}}=\frac{z}{\frac{1}{3}}$.

Hence normal is perpendicular to it.

$$\therefore (3+2\lambda) \cdot 1 + (-4+2\lambda) \cdot \frac{1}{2} + (5-3\lambda) \cdot \frac{1}{3} = 0.$$

$$\text{or } (3-2+\frac{5}{3})+\lambda(2+1-1)=0, \frac{8}{3}+2\lambda=0; \therefore \lambda=-\frac{4}{3}.$$

Putting the value of λ in (1), we get the plane as

$$x-20y+27z-14=0.$$

Ex. 6. Find the equation of the plane through the line $P \equiv ax+by+cz+d=0$, $Q \equiv a'x+b'y+c'z+d'=0$ and parallel to the line $\frac{x}{l}=\frac{y}{m}=\frac{z}{n}$.

[Kanpur 79, Meerut 73, Luck. 74, Delhi Hons. 63]

Equation of a plane through the line, i.e. the intersection of the given planes is $P+\lambda Q=0$ and d.c.'s of the normal are $a+\lambda a'$, $b+\lambda b'$, $c+\lambda c'$ and since it is parallel to the given line, the normal is perpendicular to it and hence; we have

$$l(a+\lambda a')+m(b+\lambda b')+n(c+\lambda c')=0.$$

$\therefore \lambda = -\frac{al+bm+cn}{a'l+b'm+c'n}$ and hence the required equation of the plane is $P(a'l+b'm+c'n)=Q(al+bm+cn)=0$.

Ex. 7. Find the equation of the plane through the points $(2, -1, 0)$, $(3, -4, 5)$ and parallel to the line $2x=3y=4z$.

[Vikram 70]

Any plane through $(2, -1, 0)$ is

$$\therefore A(x-2) + B(y+1) + C(z-0) = 0. \quad \dots(1)$$

It passes through $(3, -4, 5)$

$$A - 3B + 5C = 0 \quad \dots(2)$$

It is parallel to $2x = 3y = 4z$ or $\frac{x}{\frac{1}{2}} = \frac{y}{\frac{1}{3}} = \frac{z}{\frac{1}{4}}$

$\therefore$ normal is perpendicular to it.

$$\therefore A \cdot \frac{1}{2} + B \cdot \frac{1}{3} + C \cdot \frac{1}{4} = 0 \text{ or } 6A + 4B + 3C = 0. \quad \dots(3)$$

Solving (2) and (3) by cross-multiplication.

$$\frac{A}{-29} = \frac{B}{27} = \frac{C}{22} \text{ or } \frac{A}{29} = \frac{B}{-27} = \frac{C}{-22}$$

Putting the proportionate values of A, B, C in (1), we get the plane as $29(x-2) - 27(y+1) - 22(z-0) = 0$
or $29x - 27y - 22z - 85 = 0.$

(b) Find the equation of the plane through $(-1, 1, 2)$ and $(1, -1, 1)$ and perpendicular to the plane

$$x + 2y + 2z = 5$$

[Calicut 84, Rohailkhand 82]

Ex. 8. (a) Find the equation of the plane through $(3, 1, -1)$ perpendicular to the line of intersection of the planes $3x + 4y + 7z + 4 = 0$ and $x - y + 2z + 3 = 0$. [Osmania 84]

If l, m, n be the d.c.'s of the line of intersection of two planes then

$$3l + 4m + 7n = 0$$

and

$$l - m + 2n = 0$$

$$\therefore \frac{l}{15} = \frac{m}{1} = \frac{n}{-7} \text{ or D. R.'s are } 15, 1, -7.$$

The plane being perpendicular to line, its normal is parallel to line and hence D.R.'s of normal are also $15, 1, -7$. Since it passes through $(3, 1, -1)$, its equation is

$$15(x-3) + 1(y-1) - 7(z+1) = 0$$

$$15x + y - 7z - 53 = 0 \quad [\text{Cor. 1, P. 38}]$$

(b) Reduce to symmetrical form the line given by the equations $x + y + z + 1 = 0$ and $4x + y - 2z + 2 = 0$. Hence find the equation to the plane through $(1, 1, 1)$ and perpendicular to the given line.

[Meerut 81, Kashmir 75]

If l, m, n be the d.c.'s of the line then it is perpendicular to normal of both the planes

$$\therefore l + m + n = 0$$

and

$$4l + m - 2n = 0$$

$$\therefore \frac{l}{3} = \frac{m}{-6} = \frac{n}{3}$$

or $\frac{l}{1} = \frac{m}{-2} = \frac{n}{1}$

Again putting $z=0$ or solving for x and y we get the point $(-1/3, -2/3, 0)$.

Hence its equation is

$$\frac{x+1/3}{1} = \frac{y+2/3}{-2} = \frac{z-0}{1} \quad \dots(1)$$

or

Any plane through $(1, 1, 1)$ is

$$A(x-1) + B(y-1) + C(z-1) = 0 \quad \dots(2)$$

It is perpendicular to line (1) and hence normal is parallel to

it so that $\frac{A}{1} = \frac{B}{-2} = \frac{C}{1} = k$ say.

Putting for A, B and C in (2) we get the required planes as $(x-1) - 2(y-1) + (z-1) = 0$ in $x-2y+z=0$

(c) Prove that the line given by the equations $4x+4y-5z-12=0=8x+12y-13z-32$ can be written as

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z}{4}$$

[Ranchi 77, Luck. 79, Kashmir 74]

Ex. 9. Prove that the lines

$$2x+3y-4z=0, 3x-4y+z=7$$

$$5x-y-3z+12=0, x-7y+5z-6=0$$

are parallel.

[Pb. 47]

If l_1, m_1, n_1 be the d.c.'s of the first line, then

$$2l_1+3m_1-4n_1=0, 3l_1-4m_1+n_1=0.$$

$$\therefore \frac{l_1}{-13} = \frac{m_1}{-14} = \frac{n_1}{-17}$$

Similarly if l_2, m_2, n_2 be the d.c.'s of 2nd line, then

$$5l_2-m_2-3n_2=0, l_2-7m_2+5n_2=0.$$

$$\therefore \frac{l_2}{-26} = \frac{m_2}{-28} = \frac{n_2}{-34}$$

$$\therefore \frac{l_1}{l_2} = \frac{m_1}{m_2} = \frac{n_1}{n_2}. \text{ Hence they are parallel.}$$

Handwritten note: $\frac{l_1}{a_1} = \frac{m_1}{b_1} = \frac{n_1}{c_1}$ is parallel

Ex. 10. Find the angle between the lines

$$3x+2y+z-5=0, x+y-2z-3=0$$

$$2x-y-z=0, 7x+10y-8z=15.$$

Proceeding as above,

$$3l_1+2m_1+n_1=0, l_1+m_1-2n_1=0$$

or $\frac{l_1}{-5} = \frac{m_1}{7} = \frac{n_1}{1}$

$$2l_2 - m_2 - n_2 = 0, 7l_2 + 10m_2 - 8n_2 = 0$$

or $\frac{l_2}{18} = \frac{m_2}{9} = \frac{n_2}{27}$ or $\frac{l_2}{2} = \frac{m_2}{1} = \frac{n_2}{3}$,

$\therefore$ D.R.'s are $-5, 7, 1; 2, 1, 3$.

$$a_1a_2 + b_1b_2 + c_1c_2 = -10 + 7 + 3 = 0.$$

Hence the lines are perpendicular.

(b) Prove that the angle between the lines

$$x - 2y + z = 0 \text{ and } x + y - z = 0$$

and

$$x + 2y - z = 0 \text{ and } 8x + 12y + 5z = 0$$

is

$$\cos^{-1} \left(\frac{-8}{\sqrt{406}} \right)$$

[Allahabad 81, Ranchi 80]

Ex. 11. Reduce the equations of the line $x = ay + b; z = cy + d$ to the symmetrical form, and hence find the condition that this line be perpendicular to the line whose equations are $x = a'y + b'; z = c'y + d'$. [Indore 83; Garhwal 83; Karnatak 71; Utkal 78]

Here we shall not follow the usual method. From the given equation, we have

$$\frac{x-b}{a} = \frac{y}{1} = \frac{z-d}{c} \text{ and } \frac{x-b'}{a'} = \frac{y}{1} = \frac{z-d'}{c'}$$

They will be perpendicular if $aa' + 1 + cc' = 0$.

Ex. 12. Show that the line

$$x + 2y - z - 3 = 0, x + 3y - z - 4 = 0,$$

is parallel to the xz -plane and find the co-ordinates of the point where it meets yz -plane.

The equation can be reduced to symmetrical form as

$$\frac{x-1}{1} = \frac{y-1}{0} = \frac{z-0}{1} = r \text{ (say).}$$

It will be parallel to xz -plane or perpendicular to y -axis whose d.c.'s are $0, 1, 0$ if $1 \cdot 0 + 0 \cdot 1 + 1 \cdot 0 = 0$, which is true.

Any point on it is $r+1, 1, r$.

It will lie on yz -plane, i.e. $x=0$, if $r+1=0$ or $r=-1$.

Hence the point is $(0, 1, -1)$.

§ 5. To find the equation of a plane through a given line whose equations are given in.

[Meerut 80]

(a) General Form

(b) Symmetrical Form.

Let the equation of the line in general form be
 $P = a_1x + b_1y + c_1z + d_1 = 0$; $Q = a_2x + b_2y + c_2z + d_2 = 0$, ... (1)
 $P + \lambda Q = (a_1x + b_1y + c_1z + d_1) + \lambda(a_2x + b_2y + c_2z + d_2) = 0$
 is satisfied by all those points which satisfy (1) and it being an equation of first degree in x, y, z represents a plane and λ will be found by an additional given condition.

(b) Let the line be $\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1}$.

Since the required plane is to pass through the given line, hence it passes through (x_1, y_1, z_1) which is a given point on the line. But the general equation of a plane through a given point is $a(x-x_1) + b(y-y_1) + c(z-z_1) = 0$.

[Cor. 1. P. 38]. Since the plane passes through the given line, its normal is perpendicular to it.

$$\therefore al_1 + bm_1 + cn_1 = 0.$$

Hence the required plane is

$$a(x-x_1) + b(y-y_1) + c(z-z_1) = 0 \text{ where } al_1 + bm_1 + cn_1 = 0$$

Cor. To find the equation of a plane through a given line and parallel to another line.

[Mysore 73; Lucknow 71; Indore 71; Karnatak 71; Jiwaji 66; Vikram 66; Magadh 77; Rohailkhand 77]

We have already proved that the equation of the plane through a given line $\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n}$ is

$$a(x-x_1) + b(y-y_1) + c(z-z_1) \quad \dots (1)$$

$$\text{where } al_1 + bm_1 + cn_1 = 0. \quad \dots (2)$$

Again if the plane be parallel to another line whose d.c.'s are l_2, m_2, n_2 then normal will be perpendicular.

$$\therefore al_2 + bm_2 + cn_2 = 0. \quad \dots (3)$$

The required equation is obtained by eliminating a, b and c from (1), (2) and (3) and is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0. \quad (\text{V. Imp.})$$

Note. If the line be $P=0, Q=0$ then the plane through the line is $(P+\lambda Q)=0$ as given by (1) above whose normal has direction ratios

$$a_1 + \lambda a_2, b_1 + \lambda b_2, c_1 + \lambda c_2.$$

If it is parallel to line

$$\frac{x}{l} = \frac{y}{m} = \frac{z}{n}$$

then normal is perpendicular to it.

$$\therefore l(a_1 + \lambda a_2) + m(b_1 + \lambda b_2) + n(c_1 + \lambda c_2) = 0 \text{ etc.}$$

[Madurai 83]

Exercise 10

Ex. 1. Show that the equation of the plane through the line $\frac{x-1}{3} = \frac{y+6}{4} = \frac{z+1}{2}$ and parallel to $\frac{x-2}{2} = \frac{y-1}{-3} = \frac{z+4}{5}$ is $26x - 11y - 17z - 109 = 0$ and show that the point $(2, 1, -4)$ lies on it. What is the geometrical relation between the two lines and the plane.

Plane through first line is

$$A(x-1) + B(y+6) + C(z+1) = 0, \quad \dots(1)$$

where

$$3A + 4B + 2C = 0.$$

It is parallel to 2nd line and hence normal is perp. to it.

$$\therefore 2A - 3B + 5C = 0. \quad \dots(3)$$

Solving (2) and (3), we get $\frac{A}{26} = \frac{B}{-11} = \frac{C}{-17}$

Putting for A, B and C in (1), the required plane as

$$26(x-1) - 11(y+6) - 17(z+1) = 0$$

or

$$26x - 11y - 17z - 109 = 0.$$

The point $(2, 1, -4)$ clearly satisfies this plane. But [it is a point which lies on the 2nd line and hence we conclude that this plane contains the 2nd line also and hence the two given lines are coplanar, i.e. they lie in the same plane.]

Ex. 2. Find the equation of the plane which contains the line $x = \frac{y-3}{2} = \frac{z-5}{3}$, and which is perpendicular to the plane

$$2x + 7y + 3z = 1.$$

Plane through given line is

$$Ax + B(y-3) + C(z-5) = 0. \quad \dots(1)$$

where

$$A.1 + B.2 + C.3 = 0 \quad \dots(2)$$

Also the plane is perpendicular to $2x + 7y - 3z = 1$.

$$\therefore A.2 + B.7 + C(-3) = 0. \quad \dots(3)$$

Solving (2) and (3), we get

$$\frac{A}{-27} = \frac{B}{9} = \frac{C}{3} \text{ or } \frac{A}{9} = \frac{B}{-3} = \frac{C}{-1}.$$

Hence the required plane is

$$9x - 3(y - 3) - 1(z - 5) = 0 \text{ or } 9x - 3y - z + 14 = 0.$$

(b) Find the equation to the plane which contains the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{4}$ and is perpendicular to the plane $x + 2y + z = 12$.

[Jiwaji 81]

Ex. 3. Find the equation of the plane which contains the two parallel lines

$$x - 4 = -\frac{1}{4}(y - 3) = \frac{1}{8}(z - 2) \text{ and } x - 3 = -\frac{1}{4}(y + 2) = \frac{1}{8}z.$$

Equation of a plane containing the first line is

$$A(x - 4) + B(y - 3) + C(z - 2) = 0, \quad \dots(1)$$

$$\text{where } A.1 + B(-4) + C(5) = 0. \quad \dots(2)$$

The second parallel line will also lie in this plane if the point $(3, -2, 0)$ on it lies on the plane.

$$\therefore -A - 5B - 2C = 0 \text{ or } A + 5B + 2C = 0. \quad \dots(3)$$

From (2) and (3) find A, B, C and put in (1) etc.

$$11x - y - 3z = 35.$$

Ex. 4. (a) Find the equation of the plane through the line

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \text{ and parallel to the co-ordinate axes.}$$

[Agra 74]

Equation of a plane through the given line is

$$A(x - 1) + B(y - 2) + C(z - 3) = 0$$

$$\text{where } 2A + 3B + 4C = 0.$$

If the plane is parallel to x -axis, i.e. $\perp$ to yz -plane, i.e. $x = 0$;

$$\text{then } A.1 + B.0 + C.0 = 0. \quad \therefore A = 0$$

$$\text{and hence } 3B + 4C = 0 \text{ or } \frac{B}{4} = \frac{C}{-3}.$$

Thus the equation of the plane is $4(y - 2) - 3(z - 3) = 0$ or $4y - 3z + 1 = 0$. Similarly the equation of planes parallel to y -axis and z -axis are respectively $2x - z + 1 = 0$ and $3x - 2y + 1 = 0$ respectively.

(b) Find the equation of the plane which passes through the z -axis and is perpendicular to the line

$$\frac{x-1}{\cos \theta} = \frac{y+2}{\sin \theta} = \frac{z-3}{0}$$

[Lucknow 80; 75; Kanpur 80]

Any plane through z -axis $\frac{x}{0} = \frac{y}{0} = \frac{z}{1}$ is

$$Ax + By + Cz = 0 \quad \dots(1)$$

$$\text{where } A.0 + B.0 + C.1 = 0 \quad \therefore C = 0$$

It is perpendicular to given line and hence normal A, B, C is parallel to it $\therefore \frac{A}{\cos \theta} = \frac{B}{\sin \theta} = \frac{C}{0} = k$ say

Hence from (1) the plane is

$$x \cos \theta + y \sin \theta = 0$$

(c) If the plane were to contain x -axis and perpendicular to line of part (b). Determine its equation :

Plane through x -axis $\frac{x}{1} = \frac{y}{0} = \frac{z}{0}$ is

$$Ax + By + Cz = 0 \text{ where } A.1 + B.0 + C.0 = 0 \text{ i.e. } A = 0$$

It is $\perp$ to given line $\therefore \frac{A}{\cos \theta} = \frac{B}{\sin \theta} = \frac{C}{0} = k \therefore C = 0$

Also $A = 0$ so that from above

$$0 = k \cos \theta \therefore k = 0 \text{ or } \cos \theta = 0.$$

But if $k = 0$ then $B = 0$. Hence all of A, B, C are zero and we do not get the plane. Therefore we choose $\cos \theta = 0$ i.e. $\theta = \pi/2$ so that $B = k \sin \theta = k$.

Putting $A = 0, B = k, C = 0$ the plane is

$$By = 0 \text{ or } ky = 0 \text{ or } y = 0$$

Ex. 5. Find the equation of the plane through the line $\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ and through the point $(\alpha', \beta', \gamma')$.

[Garhwal 81 ; 79]

Equation of the plane through the given line is

$$A(x-\alpha) + B(y-\beta) + C(z-\gamma) = 0. \quad \dots(1)$$

where

$$Al + Bm + Cn = 0. \quad \dots(2)$$

Since the plane passes through the point $(\alpha', \beta', \gamma')$ we have

$$A(\alpha' - \alpha) + B(\beta' - \beta) + C(\gamma' - \gamma) = 0. \quad \dots(3)$$

Eliminating A, B, C between (1), (3) and (2), we get the plane as

$$\begin{vmatrix} x-\alpha & y-\beta & z-\gamma \\ \alpha'-\alpha & \beta'-\beta & \gamma'-\gamma \\ l & m & n \end{vmatrix} = 0$$

Ex. 6. Find the equation to the plane containing the line

$$\frac{x-1}{-3} = \frac{y-3}{2} = \frac{z+2}{1} \text{ and the point } (0, 7, -7) \text{ and show that the line}$$

$$\frac{x}{1} = \frac{y-7}{-3} = \frac{z+7}{2} \text{ also lies in the same plane.}$$

[Avadh 80; Berhampur 76; Garhwal 76, Punjab 60]

Equation of a plane containing given line is

$$a(x+1)+b(y-3)+c(z+2)=0. \quad \dots(1)$$

where

$$a(-3)+b.2+c.1=0. \quad \dots(2)$$

Also it passes through $(0, 7, -7)$.

$$a+4b-5c=0. \quad \dots(3)$$

$\therefore$ Solving (2) and (3) for a, b and c , we get

$$\frac{a}{-10-4} = \frac{b}{1-15} = \frac{c}{-12-2}; \therefore \frac{a}{1} = \frac{b}{1} = \frac{c}{1}$$

Hence from (1) the plane becomes

$$x+1+y-3+z+2=0 \text{ or } x+y+z=0. \quad \dots(4)$$

Now the line $\frac{x}{1} = \frac{y-7}{-3} = \frac{z+7}{2}$ will lie in this plane if the point

$(0, 7, -7)$ through which it passes satisfies its equation i.e. $0+7-7=0$ which is true and also if the normal to the plane whose D.R.'s are 1, 1, 1 is perpendicular to line whose D.R.'s are 1, -3, 2 the condition for which is that $1.1+1.(-3)+1.2=0$ which is also true. Hence the second line also lies in the plane (4).

(b) Prove that the line

$$\frac{x+1}{2} = \frac{y-1}{-3} = \frac{z+5}{3} \text{ lies on the plane } x+2y+z+4=0$$

[Burdwan 79]

Proceed as above. Show that $(-1, 1, -5)$ lies on the plane and its normal is perpendicular to line.

(c) Prove that the equation of the plane which contains the line $\frac{x+1}{2} = \frac{y-1}{3} = -(z+2)$ and passes through the point $(1, 2, -1)$ is $x-y-z=0$

[Gorakhpur 81]

Ex. 7. Prove that the plane through the point (α, β, γ) and the line $x=py+q=rz+s$ is given by

$$\begin{vmatrix} x & py+q & rz+s \\ \alpha & p\beta+q & r\gamma+s \\ 1 & 1 & 1 \end{vmatrix} = 0$$

[Meerut 69; Delhi Hons. 55; Raj. 78, 67]

The given line can be written in symmetrical form as

$$\frac{x}{1} = \frac{(y+q/p)}{1/p} = \frac{(z+s/r)}{1/r}$$

Any plane through this line is

$$Ax+B(y+q/p)+C(z+s/r)=0$$

where

$$A.1+B.1/p+C.1/r=0$$

Also it passes through (α, β, γ) ,

$$\therefore A\alpha + B(\beta + q/p) + C(\gamma + s/r) = 0.$$

Eliminating A, B, C we get the plane as

$$\begin{vmatrix} x & y+q/p & z+s/r \\ \alpha & \beta+q/p & \gamma+s/r \\ 1 & 1/p & 1/r \end{vmatrix} = 0 \text{ or } \begin{vmatrix} x & py+q & rz+s \\ \alpha & p\beta+q & r\gamma+s \\ 1 & 1 & 1 \end{vmatrix} = 0.$$

We have multiplied C_2 and C_3 by p and r respectively.

Ex. 8. Find the equation of the plane through the lines

$$ax+by+cz=0, a'x+b'y+c'z=0,$$

$$\alpha x+\beta y+\gamma z=0, \alpha'x+\beta'y+\gamma'z=0$$

[Rohailkhand 83, Meerut 72]

Both [the above given lines pass through origin and their symmetrical form is clearly seen to be

$$\frac{x}{bc'-b'c} = \frac{y}{ca'-c'a} = \frac{z}{ab'-a'b} \quad \dots(1)$$

$$\frac{x}{\beta\gamma'-\beta'\gamma} = \frac{y}{\gamma\alpha'-\gamma'\alpha} = \frac{z}{\alpha\beta'-\alpha'\beta} \quad \dots(2)$$

Plane containing (1) is

$$Ax+By+Cz=0,$$

where $A(bc'-b'c)+B(ca'-c'a)+C(ab'-a'b)=0.$

Also it contains line (2) for which two conditions are required.

I. The line (2) passes through origin ; so the plane should also pass through origin which is true, and II. Normal should be perpendicular to the line.

$$\therefore A(\beta\gamma'-\beta'\gamma)+B(\gamma\alpha'-\gamma'\alpha)+C(\alpha\beta'-\alpha'\beta)=0.$$

Eliminating A, B, C the equation of the plane is

$$\begin{vmatrix} x & y & z \\ bc'-b'c & ca'-c'a & ab'-a'b \\ \beta\gamma'-\beta'\gamma & \gamma\alpha'-\gamma'\alpha & \alpha\beta'-\alpha'\beta \end{vmatrix}$$

Ex. 9. The plane $ax+by=0$ is rotated about its line of intersection with the plane $z=0$ through an angle α . Prove that the equation of the plane in its new position is

$$ax+by \pm z\sqrt{(a^2+b^2)} \tan \alpha = 0.$$

[Allahabad 76 ; Rohailkhand 80 ; Bundelkhand 78 ; Agra 61]

The equation of the plane through the line of intersection of plane $ax+by=0$ and $z=0$ is $ax+by+\lambda z=0$. (5)

The plane is inclined to its original position i.e. to the plane $ax+by=0$ at an angle α .

$$\therefore \cos \alpha = \frac{a \cdot a + b \cdot b + \lambda \cdot 0}{\sqrt{(a^2+b^2+\lambda^2)} \cdot \sqrt{(a^2+b^2+0^2)}} = \frac{\sqrt{(a^2+b^2)}}{\sqrt{(a^2+b^2+\lambda^2)}}$$

$$\text{or } \sec^2 \alpha = \frac{a^2+b^2+\lambda^2}{a^2+b^2} \text{ or } (a^2+b^2)(\sec^2 \alpha - 1) = \lambda^2$$

$$\text{or } (a^2+b^2) \tan^2 \alpha = \lambda^2; \therefore \lambda = \pm \sqrt{(a^2+b^2) \tan^2 \alpha}.$$

Hence the equation of the plane in its new position is

$$ax+by \pm z\sqrt{(a^2+b^2) \tan^2 \alpha} = 0.$$

Ex. 10. Prove that the equation to the two planes inclined at an angle α to xy -plane and containing the line $y=0, z \cos \beta = x \sin \beta$ is $(x^2+y^2) \tan^2 \beta + z^2 - 2zx \tan \beta = y^2 \tan^2 \alpha$. [Kanpur 74 ; Agra 57]

Equation of a plane containing the given line is $(x \sin \beta - z \cos \beta) + \lambda y = 0$ and the other plane is xy -plane i.e. $z=0$ i.e. $0 \cdot x + 0 \cdot y + 1 \cdot z = 0$.

The D.R.'s of the two planes are $\sin \beta, \lambda, -\cos \beta$ and $0, 0, 1$ any they include an angle α .

$$\therefore \cos \alpha = \frac{-\cos \beta}{\sqrt{(\sin^2 \beta + \lambda^2 + \cos^2 \beta)}} \text{ or } (\lambda^2 + 1) \cos^2 \alpha = \cos^2 \beta$$

$$\therefore \lambda = \pm \frac{\sqrt{(\cos^2 \beta - \cos^2 \alpha)}}{\cos \alpha} \dots(1)$$

The two values of λ are equal and opposite.

Hence the combined equation of the two planes is

$$[(x \sin \beta - z \cos \beta) + \lambda y][(x \sin \beta - z \cos \beta) - \lambda y] = 0$$

$$\text{or } (x \sin \beta - z \cos \beta)^2 - \frac{\cos^2 \beta - \cos^2 \alpha}{\cos^2 \alpha} y^2 = 0 \text{ from (1)}$$

$$x^2 \sin^2 \beta + z^2 \cos^2 \beta - 2zx \sin \beta \cos \beta + y^2 = y^2 \cos^2 \beta \sec^2 \alpha.$$

Dividing through by $\cos^2 \beta$, we get

$$x^2 \tan^2 \beta + z^2 - 2zx \tan \beta + y^2 \sec^2 \beta = y^2 \sec^2 \alpha$$

$$\text{or } (x^2 + y^2) \tan^2 \beta + z^2 - 2zx \tan \beta + y^2 (\sec^2 \beta - \tan^2 \beta) = y^2 \sec^2 \alpha.$$

$$\text{or } (x^2 + y^2) \tan^2 \beta + z^2 - 2zx \tan \beta = y^2 (\sec^2 \alpha - 1) = y^2 \tan^2 \alpha.$$

$$\therefore \sec^2 \beta - \tan^2 \beta = 1 \text{ and } \sec^2 \alpha - 1 = \tan^2 \alpha.$$

(b) Prove that the equation of any plane through the line

$$\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n} \text{ is } (x-\alpha) \frac{\lambda}{l} + (y-\beta) \frac{\mu}{m} + (z-\gamma) \frac{\nu}{n} = 0 \text{ where}$$

$$\lambda + \mu + \nu = 0.$$

Hence find the equation of the plane containing one given line and parallel to another given line.

The given plane passes through (α, β, γ) a point of the line. Also its normal $\lambda/l, \mu/m, \nu/n$ is perpendicular to line as $l.\lambda/l + m.\mu/m + n.\nu/n = \lambda + \mu + \nu = 0$ given. Hence proved.

If the plane is parallel to some other line say

$x/l' = y/m' = z/n'$ then normal $\lambda/l, \mu/m, \nu/n$ is again perpendicular to it.

$$\therefore \frac{\lambda}{l} l' + \frac{\mu}{m} m' + \frac{\nu}{n} n' = 0.$$

Also $\lambda + \mu + \nu = 0$ solve

$$\therefore \frac{\lambda}{m' - n} = \frac{\mu}{n - l'} = \frac{\nu}{l' - m'}$$

or
$$\frac{\lambda}{mn' - m'n} = \frac{\mu}{nl' - n'l} = \frac{\nu}{lm' - l'm} = k \text{ say.}$$

Putting for λ, μ, ν in the equation of the plane and cancelling the common factor $\frac{k}{lmn}$ we get its equation as

$$(x - \alpha)(mn' - m'n) + (y - \beta)(nl' - n'l) + (z - \gamma)(lm' - l'm) = 0.$$

Foot of perpendicular from a point to a line.

Ex. 11. The axes being rectangular, find the equation to the perpendicular from the origin to the line $x + 2y + 3z + 4 = 0$; $2x + 3y + 4z + 5 = 0$; find also the co-ordinates of the foot of the perpendicular.

1st Method : Line in general form.

The perpendicular from any point P to the given line is the intersection of the following planes ;—(i) the plane through P and the given line, (ii) The plane through P perpendicular to the given line.

[Remember carefully]

The d.c.'s of the given line which is given in general form are given by

$$l + 2m + 3n = 0 \text{ and } 2l + 3m + 4n = 0 ;$$

$$\therefore \frac{l}{-1} = \frac{m}{2} = \frac{n}{-1} \text{ or } \frac{l}{1} = \frac{m}{-2} = \frac{n}{1}$$

or Its direction-ratios are $(1, -2, 1)$.

Now the plane through the origin and containing the given line is $(x + 2y + 3z + 4) + \lambda(2x + 3y + 4z + 5) = 0$.

Since it passes through origin, $\therefore \lambda = -\frac{4}{8}$, and hence the required plane is

$$3x + 2y + z = 0. \quad \dots(1)$$

Again the plane through origin and perpendicular to the given line, i.e. the line will be normal to the plane whose d.c's will be same as those of the given line, i.e. 1, -2, 1. Hence its equations is

$$1 \cdot (x-0) - 2(y-0) + 1(z-0) = 0$$

$$x - 2y + z = 0. \quad \dots(2)$$

or Equations (1) and (2) together represent the equations to the perpendicular. The d.c's of the line represented by them can as above be shown to be equal to 2, -1, -4 and both the planes pass through P, i.e. origin and hence its equation in symmetrical form is

$$\frac{x}{2} = \frac{y}{-1} = \frac{z}{-4} = r \text{ say.}$$

Foot of the perpendicular.

Any point on the perpendicular is

$$2r, -r, -4r. \quad \dots(3)$$

If it is foot of the perpendicular, then it should satisfy any of the given planes representing the given line. Putting it in $x+2y+3z+4=0$, we get $2r-2r-12r+4=0$ or $r=\frac{1}{3}$.

$\therefore$ co-ordinates of the foot of the perpendicular are

$$\left(\frac{2}{3}, -\frac{1}{3}, -\frac{4}{3}\right).$$

Note. In case the line be given in symmetrical form say $\frac{x}{2} = \frac{y}{-3} = \frac{z}{6}$, then it can be reduced to general form as $3x+2y=0$ and $2y+z=0$ and we can proceed as above.

2nd Method. Line in symmetrical form.

The given line can be reduced to symmetrical form (§ 3, P. 81) as

$$\frac{x-2}{1} = \frac{y+3}{-2} = \frac{z-0}{1} = r \text{ say.}$$

Any point on the line is N ($r+2, -2r-3, r$). If it is foot of perpendicular from P (0, 0, 0) on the given line, then PN is perpendicular to given line. D.R.'s of PN are $r+2-0, -2r-3-0, r-0$, i.e. $r+2, -2r-3, r$.

Also D.R.'s of given line are 1, -2, 1.

The condition of perpendicularity gives

$$1 \cdot (r+2) - 2 \cdot (-2r-3) + 1 \cdot r = 0$$

$$6r+8=0; \therefore r = -\frac{4}{3}$$

or

Putting the value of r , we get the foot of the perpendicular N as $\left(\frac{2}{3}, \frac{-1}{3}, \frac{-4}{3}\right)$.

Also the D.R.'s of PN are $\frac{2}{3}, \frac{-1}{3}, \frac{-4}{3}$ or $2, -1, -4$. Hence the equation to the perpendicular from origin are

$$\frac{x}{2} = \frac{y}{-1} = \frac{z}{-4}$$

Ex. 11. (b) Find the equation of the perpendicular from $(3, -1, 11)$ to the line $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ [Punjab 56 (S)]

Let the given point $(3, -1, 11)$ be P and N any point on the line be taken as $2r, 3r+2, 4r+3$.

D.R.'s of PN are $2r-3, 3r+3, 4r-8$.

If N be foot of perpendicular from P on the line then PN is perpendicular to given line. The condition of perpendicularity gives

$$2(2r-3) + 3(3r+3) + 4(4r-8) = 0 \text{ or } 29r - 29 = 0 \therefore r = 1$$

Hence the point N is $(2, 5, 7)$ and D.R.'s of PN are $-1, 6, -4$ or $1, -6, 4$. Therefore the equation to the perpendicular PN is

$$\frac{x-3}{1} = \frac{y+1}{-6} = \frac{z-11}{4}$$

Ex. 12. (a) Find the equations of the perpendicular from the point $(1, 6, 3)$ to the line $x = \frac{y-1}{2} = \frac{z-1}{3}$. Determine also the coordinates of the foot of perpendicular. [Magadh 75]

$$\text{Here } r=1, N \text{ is } (1, 3, 5) \quad \frac{x-1}{0} = \frac{y-6}{-3} = \frac{z-3}{2}$$

(b) Find the equation of the perpendicular from the point $(2, 4, -1)$ to the line $\frac{x+5}{1} = \frac{y+3}{4} = \frac{z-6}{-9}$. What will be the coordinates of the foot of perpendicular. [Gorakhpur 73]

$$\text{Here also } r=1, N(-4, 1, -3), \quad \frac{x-2}{6} = \frac{y-4}{3} = \frac{z+1}{2}$$

(c) Find the distance of the point $A(1, -2, 3)$ from the line PQ drawn through the point $P(2, -3, 5)$ making equal angles with the axes. [Barhampur 76]

Line making equal angles will have D.R.'s 1, 1, 1.

$$\therefore PQ \text{ is } \frac{x-2}{1} = \frac{y+3}{1} = \frac{z-5}{1} = r \text{ say.}$$

Rest as above $r = -2/3$, N is $(4/3, -11/3, 13/3)$.

$$AN \text{ by distance formula is } \sqrt{\left(\frac{14}{3}\right)^2}$$

Note. By distance of A from the line we mean perpendicular distance AN (N being foot of perpendicular).

Ex. 13. Find the equations of the perpendicular from origin to the line.

$$ax + by + cz + d = 0, a'x + b'y + c'z + d' = 0.$$

As discussed on P. 96 the perpendicular from P to a given line is the line of intersection plane through P and perpendicular to a given line and a plane through P and the given line. These two planes together give the equations to the perpendicular.

Any plane through given line by $P + \lambda Q = 0$ is

$$(ax + by + cz + d) + \lambda (a'x + b'y + c'z + d') = 0.$$

If it passes through P , i.e. $(0, 0, 0)$ then $d + \lambda d' = 0$

$$\text{or } \lambda = -\frac{d}{d'}$$

Hence on putting for λ the equation of the plane is

$$(ad' - a'd)x + (bd' - b'd)y + (cd' - c'd)z = 0.$$

Again D.R.'s of the given line are

$$bc' - b'c, ca' - c'a, ab' - a'b.$$

Now a plane perpendicular to given line will have its normal parallel to it. Hence the plane through $P(0, 0, 0)$ and perpendicular to given line is

$$(bc' - b'c)x + (ca' - c'a)y + (ab' - a'b)z = 0.$$

Planes (1) and (2) together give the required equations to the line.

Ex. 14. The equations to AB referred to rectangular axes are $\frac{x}{2} = \frac{y}{-3} = \frac{z}{6}$. Through a point $P(1, 2, 5)$ PN is drawn perpendicular to AB and PQ is drawn parallel to the plane $3x + 4y + 5z = 0$ to meet AB in Q . Find the equations to PN and PQ and the co-ordinates of N and Q .

Do it yourself by using 1st method. By alternative method it is done below.

Any point N on the given line is $(2r-3r, 6r)$ and P is $(1, 2, 5)$; $\therefore$ D.R.'s of PN are $(2r-1, -3r-2, 6r-5)$. PN is perpendicular to given line; therefore

$$2(2r-1) - 3(-3r-2) + 6(6r-5) = 0$$

$$\text{or } 49r - 26 = 0. \quad \therefore r = \frac{26}{49}.$$

$\therefore N$ is $(\frac{52}{49}, \frac{26}{49}, \frac{156}{49})$ and D.R.'s of PN are $\frac{3}{49}, -\frac{176}{49}, -\frac{89}{49}$ or $-3, 176, 89$; $\therefore$ equations to PN which passes through $P(1, 2, 5)$ are

$$\frac{x-1}{-3} = \frac{y-2}{176} = \frac{z-5}{89}.$$

Again point Q on the given line is

$$(2r, -3r, 6r) \text{ and } P \text{ is } (1, 2, 5),$$

D.R.'s of PQ are $2r-1, -3r-2, 6r-5$.

PQ is parallel to plane $3x+4y+5z=0$ or perpendicular to normal whose D.R.'s are $3, 4, 5$.

$$\therefore 3(2r-1) + 4(-3r-2) + 5(6r-5) = 0 \quad \therefore r = \frac{3}{2}.$$

Hence the point Q is $(3, -\frac{9}{2}, 9)$ and equations to PQ whose D.R.'s on putting for r in (1) are $4, -13, 8$ are

$$\frac{x-1}{4} = \frac{y-2}{-13} = \frac{z-5}{8}.$$

Ex. 15. The equations to AB are $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$. Through a point

$P(1, 2, 3)$ PN is drawn perpendicular to AB and PQ is drawn parallel to the plane $2x+3y+4z=0$ to meet AB in Q . Find the equations of PN and PQ and the co-ordinates of N and Q .

$$\text{Ans. } PN \text{ is } \frac{x-1}{1} = \frac{y-2}{5} = \frac{z-3}{3} \text{ and } N \left(\frac{3}{7}, \frac{-6}{7}, \frac{9}{7} \right)$$

$$PQ \text{ is } \frac{x-1}{3} = \frac{y-2}{-14} = \frac{z-3}{9}; Q \text{ is } \left(\frac{5}{2}, -5, \frac{15}{2} \right)$$

Ex. 16. Find the distance of the point $(3, 8, 2)$ from the line

$$\frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{2} \text{ measured parallel to the plane}$$

$$3x+2y-2z+15=0.$$

Let P be the given point $(3, 8, 2)$ and Q be the point on the given line whose co-ordinates are

$$(2r+1, 4r+3, 3r+2)$$

D.R.'s of PQ are $2r+1-3, 4r+3-8, 3r+2-2$

or

$$2r-2, 4r-5, 3r.$$

Now PQ is parallel to the plane $3x+2y-2z+15=0$ and hence it is perpendicular to normal $3, 2, -2$.

$$\therefore 3(2r-2)+2(4r-5)-2(3r)=0; \therefore r=2.$$

Hence the point Q is $(5, 11, 8)$ and P is $(3, 8, 2)$

$$\therefore PQ^2 = \sqrt{(2^2+3^2+6^2)}=7.$$

Projection of a line on a plane, line in sym. form.

Ex. 17. If L is the line $\frac{x-1}{2} = \frac{y}{-1} = \frac{z+2}{1}$, find the direction cosines of the projection of L on the plane $2x+y-3z=4$ and the equation of the plane through L parallel to the line $2x+5y+3z=4$, $x-y-5z=6$. [Punjab 59]

Def. The projection of a line on a given plane is the line of intersection of the given plane with another plane through given line and perpendicular to given plane.

Any plane through given line is

$$A(x-1)+By+C(z+2)=0 \quad \dots(1)$$

where

$$2A-B+C=0. \quad \dots(2)$$

This plane is perpendicular to $2x+y-3z=4$ -

$$2A+B-3C=0$$

$$\text{Solving (2) and (3), } \frac{A}{2} = \frac{B}{8} = \frac{C}{4} \text{ or } \frac{A}{1} = \frac{B}{4} = \frac{C}{2}.$$

Hence the plane (1) becomes

$$1(x-1)+4(y)+2(z+2)=0$$

or

$$x+4y+2z+3=0. \quad \dots(4)$$

This plane together with $2x+y-3z=4$(5)

gives the required equations of the projection.

If l, m, n , be the d.c.'s of the line given by (4) and (5), then $l+4m+2n=0$ and $2l+m-3n=0$.

$$\therefore \frac{l}{14} = \frac{m}{-7} = \frac{n}{7} \therefore \frac{l}{2} = \frac{m}{-1} = \frac{n}{1}.$$

$$\therefore \text{D.R.'s are } 2, -1, 1 \text{ or d.c.'s are } \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{1}{\sqrt{6}}.$$

2nd Part. D.C.'s of line given by

$$2x+5y+3z=4 \text{ and } x-y-5z=6$$

are given by $2l+5m+3n=0$, and $l-m-5n=0$.

$$\frac{l}{22} = \frac{m}{-13} = \frac{n}{7}$$

Now any plane through line L is (1) supported by (2). Also it is parallel to line whose D.R.'s are given by (6)

Normal will be perpendicular to it.

$$22A - 13B + 7C = 0 \quad \dots(7)$$

Solving (2) and (7) for A , B , and C , we get

$$\frac{A}{-7+13} = \frac{B}{22-14} = \frac{C}{-26+22} \text{ or } \frac{A}{3} = \frac{B}{4} = \frac{C}{-2}$$

Hence the required plane is

$$3(x-1) + 4(y) - 2(z+2) = 0 \text{ or } 3x + 4y - 2z = 7.$$

Line in General Form.

Ex. 18. Find the equations of the straight line which is the projection on the plane $3x + 2y + z = 0$, of the line of intersection of the planes $3x - y + 2z - 1 = 0$, $x + 2y - z = 2$. [Pb. 56]

Here the line is given in general form. We can either reduce it to symmetrical form and proceed as in Q. 17. Alternative method is to find the plane through the given line and perpendicular to given plane on which we are to find the projection. This plane together with the given plane will give us the line of projection.

Any plane through given line by $P + \lambda Q = 0$ is

$$(3x - y + 2z - 1) + \lambda(x + 2y - z - 2) = 0$$

$$\text{or } (3 + \lambda)x + (-1 + 2\lambda)y + (2 - \lambda)z - (1 + 2\lambda) = 0. \quad \dots(1)$$

It will be perpendicular to $3x + 2y + z = 0$ if

$$3 \cdot (3 + \lambda) + 2(-1 + 2\lambda) + (2 - \lambda) = 0 \text{ or } \lambda = -\frac{3}{2}.$$

Putting $\lambda = -\frac{3}{2}$ in (1), we get the plane as

$$3x - 8y + 7z + 4 = 0. \quad \dots(2)$$

The plane (2) together with the given plane

$$3x + 2y + z = 0 \quad \dots(3)$$

gives the equation to the line of projection and it can as usual be reduced to symmetrical form as

$$\frac{x + \frac{4}{5}}{11} = \frac{y - \frac{2}{5}}{-9} = \frac{z}{-15}$$

Another Definition.

Ex. 19. What do you understand by the projection of a line on a given plane? Find the equations of the projection of the line

$$\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{4} \text{ on the plane } x + 2y + z = 6.$$

Def. If P be the point of intersection of the given line and the plane and Q be the foot of the perpendicular from any point on the line to the plane, then PQ is called the projection of the given line on the given plane.

Any point on the line is $(2r+1, -r-1, 4r+3)$.

It will lie on $x+2y+z=6$ if

$$2r+1-2r-2+4r+3=6; \therefore r=1.$$

Hence the point of intersection of line and plane, i.e. P is $(3, -2, 7)$.

Foot of perp. Q from $(1, -1, 3)$ any point on the line to the plane. Any line through $(1, -1, 3)$ and perpendicular to plane $x+2y+z=6$, i.e. parallel to normal whose D.R.'s are 1, 2, 1 is

$$\frac{x-1}{1} = \frac{y+1}{2} = \frac{z-3}{1} = r \text{ say.}$$

Any point on it is $r+1, 2r-1, r+3$.

It will lie on $x+2y+z=6$ if

$$r+1+4r-2+r+3=9, \therefore r=2/3,$$

$\therefore$ foot of perpendicular Q is $(5/3, 1/3, 11/3)$

Equations to PQ joining $(3, -2, 7)$ and $(5/3, 1/3, 11/3)$

are

$$\frac{x-3}{3-\frac{5}{3}} = \frac{y+2}{-2-\frac{1}{3}} = \frac{z-7}{7-\frac{11}{3}} \quad (\text{Cor. P. 73});$$

or

$$\frac{x-3}{4} = \frac{y+2}{-7} = \frac{z-7}{10}.$$

Ex. 16. Find the equations of the line drawn through the point $(3, -4, 1)$ parallel to the plane $2x+y-z=5$ so as to cut the line $\frac{x-3}{2} = \frac{y+1}{-3} = \frac{z-2}{-1}$. Find also the co-ordinates of the point of intersection and the equation of the plane through given line and the required line.

[Punjab 52 (S)]

Let the line through $P(3, -4, 1)$ cut the given line in Q whose co-ordinates are $(2r+3, -3r-1, -r+2)$.

D.R.'s of PQ are $2r, -3r+3, -r+1$.

But PQ is parallel to the plane $2x+y-z=5$ and hence perpendicular to its normal whose D.R.'s are 2, 1, -1.

$$2(2r)+1(-3r+3)-1(-r+1)=0; \therefore r=-1.$$

Hence the point of intersection Q is $(1, 2, 3)$ and the D.R.'s of PQ are -2, 6, 2 or -1, 3, 1 and it passes through $P(3, -4, 1)$. Therefore its equation is

$$\frac{x-3}{-1} = \frac{y+4}{3} = \frac{z-1}{1} \quad \dots(1)$$

Again the equation of plane through given line and required line PQ is by § 6 P. 105.

$$\begin{vmatrix} x-3 & y+1 & z-2 \\ 2 & -3 & -1 \\ -1 & 3 & 1 \end{vmatrix} = 0.$$

§ 6. Coplanar lines.

Both lines in Symmetrical Form.

(a) To find the condition that two lines whose equations are given in symmetrical form may intersect and the equation of the plane in which they lie.

[Ravi Shankar 70; Rewa 71; Osmania 83; Burdwan 83; Vikram 72; Raj. 84; Luck 75; Calicut 87; Cal. Hons 74; Utkal 72; Avadh 82; Kanpur 81; Magadh 77; Allahabad 80; Rajasthan 76]

(a) Both Lines in symmetrical form.

Let the equation of the two lines be

$$\frac{x-x_1}{l_1} = \frac{y-y_1}{m_1} = \frac{z-z_1}{n_1} \quad \dots(1)$$

and
$$\frac{x-x_2}{l_2} = \frac{y-y_2}{m_2} = \frac{z-z_2}{n_2} \quad \dots(2)$$

The equation of a plane through the line (1) is

$$a(x-x_1)+b(y-y_1)+c(z-z_1)=0, \quad \dots(3)$$

where $al_1+bm_1+cn_1=0$ (§ 5 P. 89). $\dots(4)$

If the line (2) lies in the same plane (which would mean that the two lines intersect), then the line being perpendicular to the normal of the plane (3), we have

$$al_2+bm_2+cn_2=0 \quad \dots(5)$$

Also the point (x_2, y_2, z_2) through which the line (2) passes lies in the plane

$$\therefore a(x_2-x_1)+b(y_2-y_1)+c(z_2-z_1)=0. \quad \dots(6)$$

Hence the required condition is obtained by eliminating a, b and c between (6), (4) and (5) and is

$$\begin{vmatrix} x_2-x_1 & y_2-y_1 & z_2-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0.$$

The equation of the plane in which these lines intersect is obtained by eliminating a, b, c between (3), (4), and (5), which therefore is

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0.$$

This is also the equation of the plane through the line (1) and parallel to line (2) [§ 5]. Hence in order to find the equation of a plane in which two given lines intersect, we should write the equation of a plane through a given line and parallel to the other and in order to show that the two lines lie in the same plane, i.e. they intersect, we should show that the above plane is satisfied by the point through which the other line passes.

Alternative method :—(When point of intersection is required).

The general co-ordinates of the points on the two lines can be put in the form

$$(l_1 r_1 + x_1, m_1 r_1 + y_1, n_1 r_1 + z_1)$$

$$\text{and} \quad (l_2 r_2 + x_2, m_2 r_2 + y_2, n_2 r_2 + z_2).$$

In case the given lines intersect, then for some values of r_1 and r_2 these two points should coincide. Hence we have

$$l_1 r_1 + x_1 = l_2 r_2 + x_2 \text{ etc.}$$

$$\begin{aligned} \text{or} \quad & \left. \begin{aligned} (x_1 - x_2) + l_1 r_1 - l_2 r_2 &= 0 \\ (y_1 - y_2) + m_1 r_1 - m_2 r_2 &= 0 \\ (z_1 - z_2) + n_1 r_1 - n_2 r_2 &= 0 \end{aligned} \right\} \dots (A) \end{aligned}$$

Eliminating r_1, r_2 from the above relations we have the required condition of intersection as

$$\begin{vmatrix} x_1 - x_2 & l_1 & -l_2 \\ y_1 - y_2 & m_1 & -m_2 \\ z_1 - z_2 & n_1 & -n_2 \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ l_1 & m_1 & n_1 \\ l_2 & m_2 & n_2 \end{vmatrix} = 0$$

In-practice, however, we should solve any two of the equations (A) and find the values of r_1 and r_2 . If these values satisfy the third equation, then the lines intersect. The advantage of this method is that having found r_1 and r_2 we can also find the co-ordinates of the point of intersection of the two lines.

(b) One line in symmetrical form and other in general form.

(Allahabad 81 ; Bhagalpur 83 ; Indore 71 ;
Rajputana 75 ; Agra 73)

Find the condition that the lines

$$\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$$

and $ax+by+cz+d=0=a'x+b'y+c'z+d'$

are coplanar. Also find the equation of the plane in which they lie.

1st Method. If the two lines are coplanar, then the point on the given line should satisfy both the equations of the second line. General co-ordinates of a point on the first line are

$(lr+x_1, mr+y_1, nr+z_1)$. This point lies on

$$ax+by+cz+d=0 \text{ if } a(lr+x_1)+b(mr+y_1)+c(nr+z_1)+d=0.$$

i.e.
$$r = -\frac{ax_1+by_1+cz_1+d}{al+bm+cn}.$$

Similarly the point lies on the second plane if

$$r = -\frac{a'x_1+b'y_1+c'z_1+d'}{a'l+b'm+c'n}.$$

Equating the values of r , we have the required condition as

$$\frac{ax_1+by_1+cz_1+d}{al+bm+cn} = \frac{a'x_1+b'y_1+c'z_1+d'}{a'l+b'm+c'n}.$$

(c) Both lines in general form. Find the condition that the lines

$$a_1x+b_1y+c_1z+d_1=0=a_2x+b_2y+c_2z+d_2$$

and

$$a_3x+b_3y+c_3z+d_3=0=a_4x+b_4y+c_4z+d_4$$

may intersect.

(Ravi Shanker 70)

If the two lines intersect, then their point of intersection lies on each of the four planes representing the two lines. Hence the required condition is obtained by eliminating (x, y, z) between these four equations and is

$$\begin{vmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \\ a_4 & b_4 & c_4 & d_4 \end{vmatrix} = 0.$$

Note. In numerical examples it will be convenient to reduce the equations to symmetrical form and then follow the method of case (a) as it will be difficult to expand the determinant of fourth order.

*

Exercise 11

Ex. 1. Show that the lines

$$\frac{x-5}{4} = \frac{y-7}{4} = \frac{z+3}{-5}, \text{ and } \frac{x-8}{7} = \frac{y-4}{1} = \frac{z-5}{3}$$

are coplanar, find their common point and the equation of the plane in which they lie. [Gauhati 74; Punjab 49]

General co-ordinates of the points on the two lines are $(4r_1+5, 4r_1+7, -5r_1-3)$ and $(7r_2+8, r_2+4, 3r_2+5)$. If the two lines intersect; then these points should coincide for some value of r_1 and r_2 and hence, we have

$$4r_1+5=7r_2+8 \text{ or } 4r_1-7r_2=3.$$

$$\text{Similarly, } 4r_1-r_2=-3 \text{ and } -5r_1-3r_2=8.$$

Solving the first two equations, we get $r_2=-1$ and $r_1=-1$ and these values of r_1 and r_2 satisfy the third equation as well and hence the two lines intersect, and putting the values of r_1 and r_2 in the general co-ordinates, the common point of intersection is $(1, 3, 2)$.

Again, the equation of the plane in which they lie is the plane containing one of them and parallel to the other and its equation is

$$\begin{vmatrix} x-5 & y-7 & z+3 \\ 4 & 4 & -5 \\ 7 & 1 & 3 \end{vmatrix} = 0 \quad [\S 6 \text{ Page } 105]$$

$$\text{or } (x-5) 17 - (y-7) 47 + (z+3) (-24) = 0$$

$$\text{or } 17x - 47y - 24z + 172 = 0.$$

Clearly the point $(8, 4, 5)$ through which the second line passes also satisfies the plane. Hence the second line also lies in this plane so that the two lines are coplanar.

Ex. 2. Show that the lines

$$\frac{x+1}{-3} = \frac{y-3}{2} = z+2 \text{ and } x = \frac{y-7}{-3} = \frac{z+7}{2}$$

intersect. Find the co-ordinates of the point of intersection and the equation to the plane containing them.

[Kanpur 81; Garhwal 82; Rohailkhand 77; Agra 79]

To prove that the lines are coplanar.

Any plane containing 1st line and parallel to 2nd is

$$\begin{vmatrix} x+1 & y-3 & z+2 \\ -3 & 2 & 1 \\ 1 & -3 & 2 \end{vmatrix} = 0$$

or $(x+1)(4+3) - (y-3)(-6-1) + (z+2)(9-2) = 0$

or $x+y+z=0.$

It clearly passes through the point $(0, 7, -7)$ of the 2nd line so that this line also lies in this plane. Hence the two lines are coplanar i.e. they intersect.

Point of intersection.

Any point on (1) is $(-3r_1-1, 2r_1+3, r_1-2);$

Any point on (2) is $(r_2, -3r_2+7, 2r_2-7).$

If the lines intersect, then these points should be same.

$$\therefore -3r_1-1=r_2 \quad \text{or} \quad 3r_1+r_2=-1. \quad \dots(1)$$

$$2r_1+3=-3r_2+7 \quad \text{or} \quad 2r_1+3r_2=4. \quad \dots(2)$$

$$r_1-2=2r_2-7 \quad \text{or} \quad r_1-2r_2=-5. \quad \dots(3)$$

Solving (1) and (2), we find that $r_1=-1, r_2=2$ and these values satisfy the relation (3). Hence the lines are coplanar. Also putting for r_1 or r_2 , we find point of intersection as $(2, 1, -3).$

Ex. 3. (a) Prove that the lines

$$\frac{x}{1} = \frac{y-2}{2} = \frac{z+3}{3} \quad \text{and} \quad \frac{x-2}{2} = \frac{y-6}{3} = \frac{z-3}{4}$$

are coplanar, and find the equation of the plane in which they lie and the point of intersection.

[Madras 83, 75; Vikram 78; Indore 79; Ranchi 74; Raj. 81]

Ans. $(2, 6, 3); x-2y+z+7=0.$

(b) Prove that the lines

$$\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4} \quad \text{and} \quad \frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$$

are coplanar and the equation of the plane in which they lie is $x-2y+z=0$, and their point of intersection is $(-1, -1, -1).$

[Venkateshwara 74; Madras 74; Vikram 72; Agra 76; Punjab 81; Ranchi 82; Utkal Hons. 81]

(c) Prove that the straight lines

$$\frac{x-1}{1} = \frac{y}{-2} = \frac{z}{1} \quad \text{and} \quad \frac{x-1}{-2} = \frac{y-2}{3} = \frac{z+2}{-1}$$

are coplanar.

*

Find their point of intersection and the equation of the plane in which they lie.

[Calicut 84]

Ex. 4. Prove that the lines

$$\frac{x-a}{a'} = \frac{y-b}{b'} = \frac{z-c}{c'} \quad \text{and} \quad \frac{x-a'}{a} = \frac{y-b'}{b} = \frac{z-c'}{c}$$

intersect and find the co-ordinates of the point of intersection and the equation of the plane in which they lie. [Rajputana 84; Agra 82]

Any point on 1st is $(a'r+a, b'r+b, c'r+c)$.

Any point on 2nd is $(ar'+a', br'+b', cr'+c')$.

The lines will intersect if these points are same.

$\therefore a'r+a=ar'+a'$ Clearly these are satisfied

$b'r+b=br'+b'$, if $r=r'=1$.

$c'r+c=cr'+c'$.

Hence the lines intersect and the point of intersection on putting $r=1$ is $(a+a', b+b', c+c')$.

Again the plane through (1) and parallel to (2) is

$$\begin{vmatrix} x-a & y-b & z-c \\ a & b & c \\ a' & b' & c' \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} x & y & z \\ a & b & c \\ a' & b' & c' \end{vmatrix} = 0$$

We have added R_2 to R_1

or $x(bc'-b'c)+y(ca'-c'a)+z(ab'-a'b)=0$.

Ex. 6. Prove that the lines

$$\frac{x-a+d}{\alpha-\delta} = \frac{y-a}{\alpha} = \frac{z-a-d}{\alpha+\delta},$$

$$\frac{x-b+c}{\beta-\gamma} = \frac{y-b}{\beta} = \frac{z-b-c}{\beta+\gamma}$$

are coplanar, and find the equation to the plane in which they lie.

[Allahabad 79; Indore 78]

The lines will be coplanar if by § 6, P. 105.

$$\begin{vmatrix} a-d-b+c & a-b & a+d-b-c \\ \alpha-\delta & \alpha & \alpha+\delta \\ \beta-\gamma & \beta & \beta+\gamma \end{vmatrix} = 0,$$

Add 3rd column to first and it becomes twice the second and hence the determinant is zero as the two columns are identical. Again the equation of the plane in which they lie is

$$\begin{vmatrix} x-a+d & y-a & z-a-d \\ x-\delta & \alpha & \alpha+\delta \\ \beta-\gamma & \beta & \beta+\gamma \end{vmatrix} = 0$$

Adding 1st and third columns and subtracting twice the 2nd, we get.

$$\begin{vmatrix} x+z-2y & y-a & z-a-d \\ 0 & \alpha & \alpha+\delta \\ 0 & \beta & \beta+\gamma \end{vmatrix} = 0.$$

or $\{\alpha(\beta+\gamma) - \beta(\alpha+\delta)\}(x+z-2y) = 0$. or $x+z-2y=0$.

Ex. 7. Prove that the straight lines

$$\frac{x}{\alpha} = \frac{y}{\beta} = \frac{z}{\gamma}, \quad \frac{x}{a\alpha} = \frac{y}{b\beta} = \frac{z}{c\gamma}, \quad \frac{x}{l} = \frac{y}{m} = \frac{z}{n}$$

will lie in one plane if

$$\frac{l}{\alpha}(b-c) + \frac{m}{\beta}(c-a) + \frac{n}{\gamma}(a-b) = 0.$$

[Kanpur 81; Garhwal 78; Sambalpur 73; Cal. 73]

The three lines are drawn from origin with given direction-cosines and they will be co-planar if they are all perpendicular to the same line through O .

If L, M, N are the d.c.'s of this line, then the condition of perpendicularity gives

$$L \cdot \alpha + M \cdot \beta + N \cdot \gamma = 0,$$

$$L \cdot a\alpha + M \cdot b\beta + N \cdot c\gamma = 0,$$

$$L \cdot l + M \cdot m + N \cdot n = 0.$$

The required condition is obtained by eliminating L, M, N and is

$$\begin{vmatrix} \alpha & \beta & \gamma \\ a\alpha & b\beta & c\gamma \\ l & m & n \end{vmatrix} = 0. \quad \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ \frac{l}{\alpha} & \frac{m}{\beta} & \frac{n}{\gamma} \end{vmatrix} = 0$$

(we have divided C_1, C_2, C_3 by α, β and γ respectively)



$$\text{or } \frac{l}{\alpha}(c-b) - \frac{m}{\beta}(c-a) + \frac{n}{\gamma}(b-a) = 0$$

$$\text{or } \frac{l}{\alpha}(b-c) + \frac{m}{\beta}(c-a) + \frac{n}{\gamma}(a-b) = 0,$$

Ex. 8. Show that the lines

$$\frac{x}{\alpha/a} = \frac{y}{\beta/b} = \frac{z}{\gamma/c}, \quad \frac{x}{\alpha} = \frac{y}{\beta} = \frac{z}{\gamma}, \quad \frac{x}{a\alpha} = \frac{y}{b\beta} = \frac{z}{c\gamma}$$

are coplanar if $a=b$ or $b=c$ or $c=a$.

All the three lines pass through origin and hence as in last question, the lines will be coplanar if

$$\begin{vmatrix} \frac{\alpha}{a} & \frac{\beta}{b} & \frac{\gamma}{c} \\ \alpha & \beta & \gamma \\ a\alpha & b\beta & c\gamma \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} \frac{1}{a} & \frac{1}{b} & \frac{1}{c} \\ 1 & 1 & 1 \\ a & b & c \end{vmatrix} = 0.$$

We have cancelled α, β, γ from columns 1, 2, 3 respectively.

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = 0 \quad \text{or} \quad \begin{vmatrix} 0 & 0 & 1 \\ a-b & b-c & c \\ a^2-b^2 & b^2-c^2 & c^2 \end{vmatrix} = 0.$$

$$(a-b)(b-c) \begin{vmatrix} 1 & 1 \\ a+b & b+c \end{vmatrix} = 0 \quad \text{or} \quad (a-b)(b-c)(c-a) = 0.$$

$$a=b \quad \text{or} \quad b=c \quad \text{or} \quad c=a.$$

Ex. 10. Prove that the lines

$$x = ay + b = cz + d$$

and $x = \alpha y + \beta = \gamma z + \delta$ are coplanar if

$$(\gamma - c)(a\beta - b\alpha) - (\alpha - a)(c\delta - d\gamma) = 0.$$

[Burdwan 87, Kanpur 82]

$$\text{Symmetrical form is } \frac{x-0}{1} = \frac{y+b/a}{1/a} = \frac{z+d/c}{1/c}$$

$$\text{and } \frac{x-0}{1} = \frac{y+\beta/\alpha}{1/\alpha} = \frac{z+\delta/\gamma}{1/\gamma}.$$

They will be coplanar if by § 6, P. 105.